



Problem Solving - Pseudocode

Introduction to Pseudocode

What is Pseudocode?

- A simplified, informal high-level description of a computer program's logic
- Uses natural language mixed with programming constructs
- Bridge between human thinking and actual code
- Language-independent way to express algorithms

Why Learn Pseudocode?

- Plan before coding
- Communicate ideas clearly
- Debug logic errors early
- Language-independent problem solving

Characteristics of Good Pseudocode

Key Features

- **Simple and Clear:** Easy to read and understand
- **Concise:** No unnecessary details
- **Language Independent:** Not tied to any specific programming language
- **Structured:** Uses proper indentation and organization
- **Action-oriented:** Uses action words (calculate, read, print)

Example of Good vs Bad:

Good: READ number

Bad: scanf("%d", &number);

Good: IF age >= 18 THEN

Bad: if (age >= 18) {

Basic Structure and Syntax

Common Keywords:

- **Input/Output:** READ, INPUT, PRINT, OUTPUT, DISPLAY
- **Assignment:** SET, $\leftarrow$, =
- **Control:** IF-THEN-ELSE, WHILE, FOR, REPEAT-UNTIL
- **Logic:** AND, OR, NOT
- **Comparison:** >, <, >=, <=, =, ≠

Basic Structure:

```
BEGIN  
  statement1  
  statement2  
  ...  
END
```

Variables and Assignment

Variable Declaration:

DECLARE variable_name AS data_type

DECLARE age AS INTEGER

DECLARE name AS STRING

DECLARE salary AS REAL

Assignment Operations:

SET variable_name ← value

SET age ← 25

SET name ← "John"

SET total ← price + tax

SET average ← sum / count

Input Operations:

READ variable_name

INPUT "Enter your age: " age

Input and Output Operations

Input Statements:

READ number

INPUT "Enter your name: " name

ACCEPT value FROM user

GET student_marks

Output Statements:

PRINT "Hello World"

DISPLAY result

OUTPUT "The sum is: " total

WRITE "Welcome " + name

SHOW calculation_result

Example:

BEGIN

INPUT "Enter two numbers: " num1, num2

SET sum $\leftarrow$ num1 + num2

PRINT "Sum is: " sum

END

Arithmetic and Logical Operations

Arithmetic Operators:

- Addition: +
- Subtraction: -
- Multiplication: *
- Division: /
- Modulus: MOD or %
- Power: ^

Logical Operators:

- AND: && or AND
- OR: || or OR
- NOT: ! or NOT

Example:

SET result $\leftarrow$ (a + b) * c

SET is_valid $\leftarrow$ (age $\geq$ 18) AND (score > 50)

SET remainder $\leftarrow$ number MOD 10

Conditional Statements - IF-THEN-ELSE

Simple IF Statement:

```
IF condition THEN  
    statement(s)
```

```
END IF
```

IF-ELSE Statement:

```
IF condition THEN  
    statement(s)
```

```
ELSE  
    statement(s)
```

```
END IF
```

Nested IF Statement:

```
IF condition1 THEN  
    IF condition2 THEN  
        statement(s)
```

```
    END IF
```

```
ELSE  
    statement(s)
```

```
END IF
```

Multiple Conditions – IF–ELSE–IF

Multi-way Selection:

```
IF condition1 THEN
    statement(s)
ELSE IF condition2 THEN
    statement(s)
ELSE IF condition3 THEN
    statement(s)
ELSE
    statement(s)
END IF
```

Example - Grade Calculator:

```
IF marks >= 90 THEN
    PRINT "Grade A"
ELSE IF marks >= 80 THEN
    PRINT "Grade B"
ELSE IF marks >= 70 THEN
    PRINT "Grade C"
ELSE
    PRINT "Grade F"
END IF
```

Loops - REPEAT-UNTIL

REPEAT-UNTIL Structure:

REPEAT

 statement(s)

UNTIL condition

Key Difference:

- Executes at least once (condition checked at end)
- Continues until condition becomes TRUE

Example - Menu System:

REPEAT

 PRINT "1. Add 2. Subtract 3. Exit"

 INPUT "Choose option: " choice

 IF choice = 1 THEN

 // Add operation

 ELSE IF choice = 2 THEN

 // Subtract operation

 END IF

UNTIL choice = 3

Complete Example - Student Grade System

```
BEGIN
  DECLARE student_name AS STRING
  DECLARE marks[5] AS INTEGER
  DECLARE total, average AS REAL
  DECLARE grade AS STRING

  // Input student details
  INPUT "Enter student name: " student_name

  // Input marks for 5 subjects
  SET total ← 0
  FOR i FROM 0 TO 4 DO
    INPUT "Enter marks for subject " (i+1) ": " marks[i]
    SET total ← total + marks[i]
  END FOR
```

Complete Example - Student Grade System

// Calculate average

SET average $\leftarrow$ total / 5

// Determine grade

IF average $\geq$ 90 THEN

SET grade $\leftarrow$ "A+"

ELSE IF average $\geq$ 80 THEN

SET grade $\leftarrow$ "A"

ELSE IF average $\geq$ 70 THEN

SET grade $\leftarrow$ "B"

ELSE IF average $\geq$ 60 THEN

SET grade $\leftarrow$ "C"

ELSE

SET grade $\leftarrow$ "F"

END IF

Complete Example - Student Grade System

```
// Display results
```

```
PRINT "Student Name: " student_name
```

```
PRINT "Total Marks: " total
```

```
PRINT "Average: " average
```

```
PRINT "Grade: " grade
```

```
END
```

Best Practices and Tips

Writing Effective Pseudocode:

- **Start with the big picture** - Outline main steps first
- **Use meaningful variable names** - student_age not sa
- **Keep it simple** - One operation per line
- **Use proper indentation** - Shows program structure clearly
- **Be consistent** - Use same style throughout
- **Add comments** - Explain complex logic
- **Test with examples** - Walk through with sample data

Common Mistakes to Avoid:

- Being too detailed (like actual code)
- Mixing different notation styles
- Poor indentation
- Unclear variable names
- Missing END statements